INCH-POUND
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PERFORMANCE SPECIFICATION

TANK, FUEL, ENGINE: NON-METALLIC, NON-SELF-SEALING, FOR MILITARY VEHICLES

This specification is approved for use within the U.S. Army Tank-automotive and Armaments Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of non-metallic, non-self-sealing fuel tanks (fuel cell type) for use in military tactical vehicles and combat vehicles.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirement documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 2910

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D381	- Existent Gum in Fuels by Jet Evaporation, Test Method for (DoD Adopted).
ASTM D412	- Rubber Properties in Tension, Test Methods for (DoD Adopted).
ASTM D413	- Rubber Property-Adhesion to Flexible Substrate, Test Methods for (DoD Adopted).
ASTM D471	- Standard Test Method for Rubber Property - Effect of Liquids (DoD Adopted).
ASTM D5486	- Standard Specification for Pressure-Sensitive Tape for Packaging, Box Closure, and Sealing (DoD Adopted).

(Copies of ASTM publications may be obtained from the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.3.

3.2 Design, materials, and manufacturing process. Unless otherwise specified (see 6.2), the design, materials, and manufacturing process selection is the prerogative of the contractor as long as all articles submitted to the government fully meet the interface and interoperability, environmental, support and ownership, and operating requirements specified herein.

3.2.1 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Interface and interoperability. The fuel tank shall be of flexible construction, and shall consist generally of a nonmetallic carcass of unit construction, with openings or fittings satisfactory for the necessary attachment of parts and accessories in accordance with applicable drawings. Fuel tanks shall conform to the envelope dimensions, contours, fitting and mounting interfaces as shown on applicable drawings (see 6.2).

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3.3.1 Installation. The location of all fittings and the tank dimensions needed for installation shall be within the allowable tolerances specified on the applicable drawings.

3.3.2 Stand. The fuel tank shall show no evidence of leakage as a result of standing.

3.3.3 Fuel contamination. The nonvolatile material, for a sample of the inner layers of fuel tank, shall not exceed 60 milligrams (mg) per 100 milliliters (ml) of the contaminated fluid. The stoved gum residue shall not exceed 20 mg per 100 ml of the contaminated fluid.

3.4 Environmental.

3.4.1 Humidity (hydrolytic stability) at elevated temperature. The specimen shall show no corrosion, peeling, cracking, warping, blistering, delamination, or discoloration as a result of exposure to high humidity at elevated temperatures.

3.4.2 Humidity (hydrolytic stability) at room temperature. The specimen shall show no visible cracks, blisters or eruptions after exposure to high humidity at room temperature.

3.4.3 Low temperature leakage. The fuel tank shall show no evidence of leakage as a result of exposure to low temperatures.

3.5 Support and ownership.

3.5.1 Hazardous materials. The fuel tank shall not produce any hazards to personnel or the environment resulting from the use of asbestos or cadmium.

3.5.2 Protective treatments.

3.5.2.1 Fittings. Fittings shall be corrosion resistant or appropriately treated for corrosion resistance.

3.5.2.2 Openings. All openings to tanks shall be fitted with fully cured Buna-N or equally fuel resistant material which shall be resistant to cutting, and to gouging as a result of sealing pressure generated by torquing hold-down bolts.

3.5.3 Access opening. Each fuel tank shall be provided with a hand hole and cover to permit arm access to the tank interior for inspection and repair.

3.5.4 Handling. The fuel tank shall show no evidence of cracks, loosening of fittings, or failure of seams when it is folded and unfolded 40 times in accordance with standard procedure for shipment.

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3.5.5 Fluid resistance of exterior surfaces. The exterior surfaces of the fuel tanks shall show no swelling, separation, blistering, or dissolution when exposed to fuel.

3.5.6 Permeability. Fuel tank permeability shall be less than 0.025 fluid ounces (fl-oz) per square foot (ft^2) per 24 hours for each specimen tested.

3.5.7 Tensile strength. The inner layer ply of the tank, without barrier, shall have a tensile strength not less than 80 percent (%) of the value calculated, based on the original cross sectional area of the ply.

3.5.8 Seam adhesion. The seam adhesion of the inner ply of the fuel tank shall be not less than 12 pounds (lb). There shall be no failure of the seam.

3.5.8.1 Wet seam adhesion. The wet seam adhesion shall be not less than 50% of the dry adhesion value.

3.5.9 Puncture resistance. The force required to puncture the fuel tank wall shall be not less than 15 lb.

3.5.10 Identification marking. Each fuel tank shall have the following information permanently and legibly marked on the face or surface in which the opening for the fuel-level gage sending unit is located:

Army part number.
Vehicle nomenclature.
Name of manufacturer.
Date of manufacture (year and quarter).
Manufacturer’s serial number.
Manufacturer’s construction number.
Specification MIL-PRF-14442.
Date of acceptance.
Contract or order number.

3.5.10.1 Filler cap. The external surface of the filler cap shall be formed and painted with diesel fuel (DF-A, DF-1, DF-2, and JP-8) resistant lacquer, and legibly marked in white lettering: “Fuel (*) US Gal.” (* to be filled in with usable capacity taken as the average capacity of the first 10 production tanks).

3.5.10.2 Hand-hole cover. The exterior surface of the hand-hole cover of each fuel tank shall be legibly and permanently marked “OUTSIDE”.

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3.5.10.3 Torque values. The torque values required for assembly shall be marked on, or adjacent to, each fitting or accessory involved, on the side torque wrench is applied. If the torque required is the same for each such item, that value may be stenciled in only one place on the surface of tank in a conspicuous location.

3.6 Operating. When installed in the vehicle structure for which it is designed, the fuel tank shall be capable of withstanding the loads incident to road shock and vibrations common to off-the-road and high speed highway operation of the vehicle.

3.6.1 Capacity. The nominal capacity of the fuel tanks shall be as specified with a tolerance of ± 3.0% (see 6.2).

3.6.2 Leakage. The fuel tank shall show no evidence of leakage.

3.6.3 Slosh. The fuel tank shall show no evidence of leakage or other failure such as chafing, disengaged hangers, or liner breaks as a result of content sloshing.

4. VERIFICATION

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Conformance inspections (CI) (see 4.4).

4.2 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be conducted under the following conditions:

a. Air temperature: 73 ± 3 degrees Fahrenheit (°F)
b. Barometric pressure: 28.5 (+2, -3) inches (in.) mercury
c. Relative humidity: 50 ± 5%

4.2.1 Test structure. When specified (see 6.2), the test structure shall be a simulation of the actual vehicle structure. It shall duplicate the shape, dimensions and material of the tank supporting structure in the vehicle. The necessary stops, cushions, and pads, identical with those used in the finished vehicle, for mounting and supporting the tank, shall be provided. It shall be constructed so that the test tank can be installed in and removed from the structure in an identical manner to that used in the actual vehicle installation.

4.2.2 Support jig. The jig shall support the tank test structure specified in 4.2.1, to simulate the actual vehicle installation. The support jig shall be a strongly constructed framework of metal, for carrying the mounted test tank and designed for bolting to the rocker assembly

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specified in 4.9.2. The jig shall be designed so that removal of the tank test structure may be accomplished with a minimum effort.

4.3 First article inspection. When a first article sample is required (see 3.1), samples shall be inspected as specified in table I. First article samples shall be selected as follows:

Three full size tanks complete with necessary fittings.
Three 12 by 12 in. specimens of composite tank construction.
One specimen of inner layer ply, without barrier, approximately 900 square inches (in^2) in area, including seam.
One specimen of 6 by 6 in. inner layer ply, without barrier.
Three permeability specimens in accordance with 4.8.3.2.

TABLE I. Classification of inspections.

Title	Requirement	Verification	Title	Requirement	Verification
Interface and interoperability	3.3	4.6	Fluid resistance of exterior surfaces	3.5.5	4.8.2
Installation	3.3.1	4.6.1	Permeability	3.5.6	4.8.3
Stand	3.3.2	4.6.2	Tensile strength	3.5.7	4.8.4
Fuel contamination	3.3.3	4.6.3	Seam adhesion	3.5.8	4.8.5, 4.8.5.1
Environmental	3.4	4.7	Wet seam adhesion	3.5.8.1	4.8.5.2
Humidity at elevated temperature	3.4.1	4.7.1	Puncture resistance	3.5.9	4.8.6
Humidity at room temperature	3.4.2	4.7.2	Identification marking	3.5.10	4.8
Low temperature leakage	3.4.3	4.7.3	Filler cap	3.5.10.1	4.8
Support and ownership	3.5	4.8	Hand-hole cover	3.5.10.2	4.8
Hazardous materials	3.5.1	4.8	Torque values	3.5.10.3	4.8
Protective treatments	3.5.2	4.8	**Operating**	3.6	4.9
Fittings	3.5.2.1	4.8	Capacity	3.6.1	4.9
Openings	3.5.2.2	4.8	Leakage	3.6.2	4.9.1
Access opening	3.5.3	4.8	Slosh	3.6.3	4.9.2
Handling	3.5.4	4.8.1			

4.4 Conformance inspection. Conformance inspection shall include the applicable tests of table I as specified in the contract (see 6.2).

4.5 Verification methods. The types of verification methods included in this section are examination of contractor records, visual inspection, measurement, sample tests, full-scale demonstration tests, simulation, modeling, engineering evaluation, component properties analysis, and similarity to previously-approved or previously-qualified designs.

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4.5.1 Verification alternatives. The manufacturer may propose alternative test methods, techniques, or equipment, including the application of statistical process control, tool control, or cost-effective sampling procedures, to verify performance.

4.6 Interface and interoperability. Conformance to 3.3 through 3.3.3 shall be determined using appropriate verification methods specified in 4.5 or 4.5.1 and the tests specified in 4.6.1 through 4.6.3.2.

4.6.1 Installation.

4.6.1.1 First article sample. For first article testing, conformance to 3.3.1 shall be determined using the test structure specified in 4.2.1.

4.6.1.2 Quality conformance. For conformance testing, conformance to 3.3.1 shall be determined, using a test jig (see 4.2.2) or registered metal templates fabricated in accordance with design requirements furnished by the vehicle manufacturer for the particular vehicle.

4.6.2 Stand. To determine conformance to 3.3.2, the tank shall be tested as specified in 4.6.2.1 or 4.6.2.2, as applicable.

4.6.2.1 First article. Upon completion of the low temperature test specified in 4.7.3, the cavity shall be completely lined with brown paper. The tank shall be refilled with fluid conforming to reference fuel B of ASTM D471. The tank shall be allowed to stand at the standard ambient temperature for a period of 30 days, at which time the paper shall be examined for any stains indicating a fluid leakage.

4.6.2.2 Conformance testing. Each tank selected for conformance inspection shall be collapsed and held for 30 minutes in a position comparable to that encountered prior to installation in its respective vehicle cavity. Prior to the installation of the tank, the test structure shall be completely lined with brown paper and held in place with tape conforming to ASTM D5486. Then the tank shall be installed in its test structure. The tank shall then be filled with fluid conforming reference fuel B of ASTM D471 and shall be stand tested for 30 days. The tank may be inspected at 10-day intervals during the test. At the conclusion of the 30-day test, the tank shall be drained and removed from the test cavity. The test cavity shall be inspected for evidence of leakage as indicated by stains on the brown paper. After the examination, if stains are present on the brown paper, evaluation shall be made in accordance with the test procedure specified in 4.9.2.

4.6.3 Fuel contamination.

4.6.3.1 Nonvolatile gum residue. A 5 gram sample of the inner layers, up to the barrier, shall be diced into approximately 0.06 in. squares and shall be placed in a flask containing 230 ml

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of fluid conforming to reference fuel B of ASTM D471. The samples shall be allowed to stand for 48 hours at 77 ± 5°F. The contaminated test fluid shall be decanted off, and the nonvolatile gum residue shall be determined by the method specified in ASTM D381. The total evaporation time shall be 45 minutes.

4.6.3.2 Stoved gum residue. Stoved gum residue content shall be determined after necessary corrections have been made for performed gums, originally present in the test fluid. The beakers containing the nonvolatile material shall be placed in an appropriate bath maintained at a constant temperature of 572 ± 9°F for 30 minutes, and then weighed after being cooled in a closed container.

4.7 Environmental.

4.7.1 Humidity (hydrolytic stability) at elevated temperature. To determine conformance to 3.4.1, one 6 by 6 in. specimen of the composite tank construction shall be examined after being exposed to a temperature of 158 ± 2°F and 100% relative humidity for a period of 15 days.

4.7.2 Humidity (hydrolytic stability) at room temperature. To determine conformance to 3.4.2, two specimens 1 by 6 by 0.075 ± 0.025 in. shall be examined under 7-power magnification after being exposed to room temperature and 100% relative humidity for 30 days. When practical, these specimens shall be mounted on an appropriate fixture in a manner to cause a stretch of 20%, measured between gage marks 4 in. apart, centered in the middle section of each strip.

4.7.3 Low temperature leakage. To determine conformance to 3.4.3, the fuel tank shall be emptied, after the test specified in 4.9.2, and the interior of the cavity in which the tank is installed shall be completely lined with brown paper. The tank shall be refilled with a test fluid conforming to reference fuel B of ASTM D471 containing a staining agent, and allowed to stand for a period of 7 days at 110 ± 10°F. The tank shall then be emptied and subjected to an air dry-out at 160 ± 10°F for 7 days. Following the air dry-out, the tank shall be refilled with the same test fluid specified and cooled to a temperature of -67 ± 3.6°F and allowed to remain at this temperature for a minimum period of 3 days. The tank shall then be brought back to room temperature and allowed to stand until completely drained and then examined for leakage as indicated by staining the brown paper.

4.8 Support and ownership. Conformance to 3.5 through 3.5.10.3 shall be determined using appropriate verification methods specified in 4.5 or 4.5.1 and the tests specified in 4.8.l through 4.8.6.

4.8.1 Handling. To determine conformance to 3.5.4, the fuel tank shall be folded and unfolded as specified and then installed and removed from the test structure five times.

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Applicable service procedures shall be followed in tank installation and removal. All fittings shall be fastened to corresponding structure and interconnect fittings upon each installation in the test structure, and the tank shall then be examined.

4.8.2 Fluid resistance of exterior surfaces. To determine conformance to 3.5.5, a 6 by 6 in. specimen of composite tank construction shall be submerged in a container of test fluid conforming to reference fuel B of ASTM D471 at standard ambient temperature for 24 hours.

4.8.3 Permeability. To determine conformance to 3.5.6, the fuel tank shall be tested as specified in 4.8.3.1 through 4.8.3.3.

4.8.3.1 Test apparatus. The test apparatus shall include a permeability cup and ring constructed in accordance with figure 1. A nylon solution shall be used for sealing the test disk to the permeability cup.

4.8.3.2 Preparation of test specimen. The uncured liner shall be applied to a 10 by 10 in. piece of corrugated fiberboard coated on one side with a suitable water soluble breakaway agent. The exposed surface of the inner liner shall be coated with prime cement and barrier resin, if required, in accordance with applicable manufacturing specifications. The assembly shall be wrapped with cellophane and covered with a waterproof bag. The assembly shall be vulcanized using the same method as in regular production. After vulcanization, the waterproof bag and cellophane shall be removed, and the inner liner shall be removed from the fiberboard, using water if necessary. Free moisture shall be wiped from the assembly, and the assembly shall be conditioned for 24 hours at a constant temperature of 77 $\pm$ 50°F and a relative humidity of 40 $\pm$ 5%. Two disks, each 2.5 in. in size, shall be cut from the vulcanized panel. One hundred ml of the test fluid conforming to reference fuel B of ASTM D471, shall be placed in the diffusion cup shown in figure 1. A nylon solution shall be applied to the face of the cup flange covering the area inside the bolt circle. The nylon solution shall be allowed to come almost to dryness, then the test disk shall be applied to the cup with the barrier, if any, facing outward. The assembly shall be completed by attaching the bolting ring and tightening the bolts in accordance with the following schedule:

Inner liner type	Bolt torque, pound-inches (lb-in)
Gum stock	5 to 10
Coated fabrics	15 to 20
Unsupported plastic films	20 to 25

4.8.3.3 Method of conducting test. Permeability cups, prepared as specified in 4.8.3.2, shall be placed in a rack in a constant temperature of 77 $\pm$ 5°F and relative humidity of 40 $\pm$ 5%. After allowing one hour for equilibration, the cup shall be weighed to the nearest 0.005 gram (g) and placed in the rack with the cup facing upward. The cup shall be kept at the above constant

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temperature for 24 hours, then weighed to check for excessive vapor loss. The bolts shall be retorqued if necessary. The cup shall be inverted (test disk down) in a rack that permits free access of air to the test disk. Cups shall be weighed at the end of the third, fifth, and eighth day after inverting. Defective films or leaks caused by faulty assembly will usually be found when weighing on the third day. The diffusion rate calculation shall be made on the fifth to the eighth day period, and the permeability shall then be determined. Diffusion expressed in fl-oz per ft^2 per 24 hours equals the gram loss of the test specimen per 24 hours multiplied by a factor K which is defined as follows:

$$K = \frac{144}{(\text{sp. gr.})\ (29.573)\ (3.1416)\ (R)^2}$$

Where:

sp. gr. = specific gravity of the test fluid at 77°F.
R = inside radius of test cup expressed in inches.

4.8.4 Tensile strength. To determine conformance to 3.5.7, the tensile strength of a sample of inner layer ply, without barrier, shall be determined in accordance with the test specified in ASTM D412, when coated fabric liners are used. The sample shall then be immersed in water at 180 ± 4°F for 72 hours, after which the specified tensile strength test shall be repeated. The change in tensile strength shall be reported as the after-immersion strength, calculated as a percentage of the original value.

4.8.5 Seam adhesion. To determine conformance to 3.5.8, the seam adhesion of the inner layer ply to itself shall be tested along the length of the seam in accordance with the test specified in ASTM D413, using the strip specimen.

4.8.5.1 Seam adhesion (alternative procedure). To determine conformance to 3.5.8, the seam adhesion of the inner layer ply to itself may be tested by cutting a strip of inner-layer material one in. wide, having a seam made in the same manner as is used in the tanks submitted under 4.3. This seam shall be perpendicular to, and midway in, the length of the strip. A tensile load shall be applied parallel to the length of the strip, of sufficient magnitude to break the strip.

4.8.5.2 Wet seam adhesion. To determine conformance to 3.5.8.1, a specimen shall be prepared in accordance with the test of 4.8.5 or 4.8.5.1, and soaked in a test fluid conforming to referenced fuel B of ASTM D471, for a period of 48 hours, at 75 ± 5°F. The specimen shall be removed from the test fluid and wiped dry. No other means of drying shall be used. Within 4 hours after removing the specimen from the fluid, it shall be subjected to one of the above tests. The value by the strip method shall be no less than 50% of the dry test value. The results using the alternative procedure shall require the breaking of the strip outside of seam area.

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4.8.6 Puncture resistance. To determine conformance to 3.5.9, a test specimen of tank wall construction shall be fastened in a specimen holder as shown in figure 2. A piercing instrument, with its piercing end as shown in figure 2, shall be forced against the specimen in the holder at approximately the center of the area enclosed by the specimen holder. The force required to puncture the specimen shall be not less than specified.

4.9 Operating. Conformance to 3.6 through 3.6.3 shall be determined using appropriate verification methods specified in 4.5 or 4.5.1 and the tests in 4.9.1 and 4.9.2.

4.9.1 Leakage. To determine conformance to 3.6.2, the tank shall be covered with brown paper and shall be filled with a stoddard solvent (or equivalent) containing a staining agent. The tank and supported systems shall be subjected to an internal pressure of 1.5 times the maximum operating pressure of the intended system and allowed to stand for 30 minutes. Tanks intended for use in non-pressurized systems shall be subjected to an internal pressure equivalent to the normal head pressure measured at the bottom of the tank multiplied by a factor of 1.5 and shall be allowed to stand for 30 minutes. The brown paper shall then be examined for stains.

4.9.2 Slosh. To determine conformance to 3.6.3, the test tank, complete with all fittings and other parts or accessories, shall be mounted in the support jig (see 4.2.2) and installed on a rocker assembly. It shall be mounted in such a manner that the major (longer) horizontal axis of the tank shall be 90 degrees (°) to the center line of the axis of the rocker assembly platform. The interior of the compartment shall be lined with brown paper in the fitting areas and held in place with tape conforming to ASTM D5486. The test specimen shall be filled two-thirds full with test fluid conforming to reference fuel B of ASTM D471. In addition, twice the sump capacity of water shall be added to the tank. The tank shall then be subjected to rocking at approximately 15° angle from the vertical (30° total rocking for 24 hours at a frequency of 16 to 20 cycles per minute (cpm)), or for 40 hours at 10 to 16 cpm. The temperature of the test fluid during the test shall be 110 $\pm$ 10°F. Tanks intended for use in pressurized fuel systems shall be subjected to an internal pressure of 1.5 times the maximum operating pressure of the intended vehicle system during the entire test. Tanks intended for use in non-pressurized fuel systems shall be tested at normal atmospheric pressure. When testing non-pressurized tanks, a vent opening shall be provided equal to the vent opening in the intended vehicle. The vent shall be connected to a line and so positioned that any test fluid escaping through the vent shall be carried away from the test specimen. At the conclusion of the 24 or 40 hours rocking period, as applicable, the tank shall be completely filled with the specified test fluid and allowed to stand for 1 hour. The test fluid shall then be drained and the test tank removed from the structure. The paper lining shall be thoroughly examined for leakage as indicated by stains on the paper. Indication of stains on the liner paper shall be considered as test failure, except minor stainage due to improperly torqued bolts, damaged equipment not considered a part of the tank assembly, or minor stainage resulting from physical damage to the tank at the time of installation.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

6.1 Intended use. The nonmetallic, non-self-sealing fuel tanks covered by this specification are intended for installation in military tactical vehicles including trucks, truck-tractors, and similar thin-skinned wheeled vehicles, as reservoirs for compression ignition fuel. They may also be installed on combat vehicles, to provide corrosion-resistant fuel storage reservoirs. The fuel tanks are intended for use with fuel which does not exceed 30% aromatics content.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2).
c. First article inspection, if required (see 3.1 and 6.6).
d. Design, materials, or manufacturing processes, if other than specified (see 3.2).
e. Title, number, and date of applicable drawings (see 3.3).
f. Capacity of the fuel tank (see 3.6.1).
g. If inspection conditions should be other than specified (see 4.2).
h. A test structure to support first article inspection, when applicable (see 4.2.1).
i. Conformance tests to be performed (see 4.4 and 6.7).
j. Selection of applicable level and packaging requirements (see 5.1).

6.3 Rubber material. Satisfactory results have been obtained with fuel tanks constructed with fabric-reinforced nitrate rubber.

6.4 Surface treatment for steel fittings. Refer to SAE AIR4160 for information regarding possible alternatives to the use of cadmium plating on fittings.

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6.5 Tanks. Following is a partial list of drawings which describe the fuel tank shells which the items covered by this specification are intended to fit.

Drawing number	Title	Vehicle applications
7993036	Gas Tank Cell, Right	M53, M55
7993037	Gas Tank Cell, Left	M53, M55
8340664	Fuel Cell Assembly	M59, M84
10918604	Tank, Fuel, Engine	M114, M114A1, M114A1E1, M114E2
10952618	Cell Assembly, Fuel, Engine (Center)	M551A1
11633639	Tank, Fuel, Engine	M132, XM806
11647693	Bladder, Fuel	M741

6.6 First article. When requiring a first article inspection, contracting documents should provide specific guidance to offerors. This guidance should cover whether the first article is a first article sample, a first production item, or the number of test items. These documents should also include specified instructions regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Pre-solicitation documents should provide Government waiver rights for samples for first article inspection to bidders offering a previously acquired or tested product. Bidders offering such products who wish to rely on such production testing must furnish evidence with the bid that prior Government approval is appropriate for the pending contract.

6.7 Conformance inspection. Affordable conformance inspection with confidence varies depending upon a number of procurement risk factors. Some of these factors include: Contractor past performance, government schedules and budget, product material and design maturity, manufacturing capital equipment and processes applied, the controlled uniformity of those processes, labor skill and training, and the uniformity of measuring processes and techniques. During the solicitation, contracting documents should indicate those tests desired from table I and their designated frequency based on a risk assessment for the procurement.

6.8 Subject term (key word) listing.

Carcass
Cell type
Diesel
Fluid container
Gasoline

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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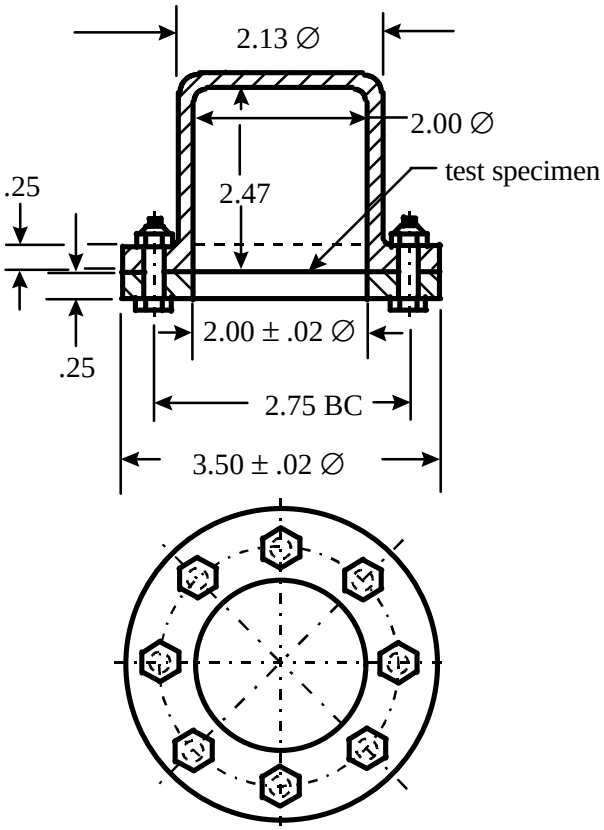


FIGURE 1. Diffusion cup assembly.

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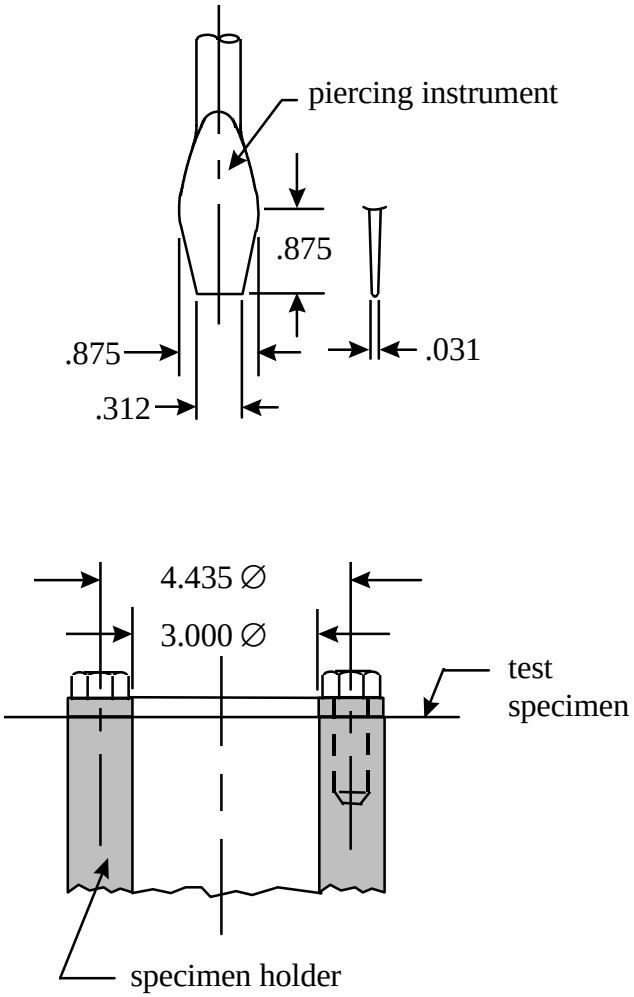


FIGURE 2. Piercing instrument and specimen holder.

Custodian:
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Preparing Activity:
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(Project 2910-A012)

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1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
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I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-PRF-14442E(AT)	**2. DOCUMENT DATE *(YYMMDD)*** 971121

3. DOCUMENT TITLE

Tank, Fuel, Engine: Non-Metallic, Non-Self-Sealing, For Military Vehicles

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	**b. ORGANIZATION**	
c. ADDRESS *(Include Zip Code)*	**d. TELEPHONE *(Include Area Code)*** **(1) Commercial** **(2) AUTOVON** ***(If applicable)***	**7. DATE SUBMITTED *(YYMMDD)***

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c. ADDRESS *(Include Zip Code)* **Commander** **U.S. Army Tank-automotive and Armaments Command** **ATTN: AMSTA-TR-E/BLUE** **Warren, MI 48397-5000**	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** **Defense Quality and Standardization Office** **5203 Leesburg Pike, Suite 1403** **Falls Church, VA 22041-3466** **Telephone (703) 756-2340 AUTOVON 289-2340**

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